

Title: Systems and Methods for Symmetry-Constrained Spatial Computation

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Attorney/Agent: Pro se (none)

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SPECIFICATION

1. Title of the Invention

Systems and Methods for Symmetry-Constrained Spatial Computation

2. Cross-Reference to Related Applications

This application claims the benefit of conceptual disclosure related to symmetry-constrained computation, volumetric logic points, and spatial logic architectures. No formal related patent applications have yet been filed.

3. Field of the Invention

This invention relates to **computing architectures** and **finite-state logic systems** that use **spatial constraints and symmetry relationships** to encode and execute logical operations. Applications include hardware, software, AI systems, robotics, biological modeling, educational systems, and knowledge representation.

4. Background of the Invention

Traditional logic systems rely on:

- Symbolic encoding of state,

- Sequential execution flows,
- Global clock synchronization,
- Wired gate interconnections.

These design constraints create significant limitations:

- **Illegal states** remain representable,
- High power consumption for constraint enforcement,
- Brittleness in mapping real-world structure to computation,
- Exponential growth in symbolic complexity.

Many systems — including biological, physical, and robotic processes — are **spatially constrained** rather than symbol-driven. A logic substrate that **makes illegal states physically impossible** can increase efficiency, safety, and interpretability.

5. Summary of the Invention

The invention provides:

1. **Symmetry-Constrained Spatial Computation (SCSC)** — A paradigm where allowable states are **geometrically encoded**, and transitions follow **topological adjacencies**.
2. **Volumetric Logic Points** — Units of information represented in **polyhedral formations** (e.g., tetrahedra, octahedra) that inherently encode:
 - Boolean values,
 - Symmetry classes,
 - Finite transformation sets.
3. **Reconfigurable Spatial Substrates** — Dynamic state architectures where the underlying adjacency network can be:
 - rearranged,
 - constrained,
 - expanded or reduced.
4. **Execution Layer** — A processor model enabling:

- parallel traversal of legal transitions,
- prevention of invalid state movement by geometry,
- Boolean completeness using spatial primitives.

This provides **integrated logical, computational, and representational power**.

6. Detailed Description of the Invention (Excerpt)

6.1 Symmetry-Constrained State Representation

Each logic element is represented as a spatial formation embedded in a finite topological structure. A state is “legal” if and only if it satisfies defined **spatial adjacency constraints**.

Example configurations include:

- **Tetrahedral encodings** → 2-bit systems
- **Octahedral or cubic formations** → multi-bit, multi-facet relationships

Illegal states lack a valid spatial embedding and are thus **physically excluded**.

6.2 Spatial Logic Operations

Operations are performed by:

1. Identifying allowed transforms based on **nearest-neighbor adjacency**.
2. Applying transitions that **preserve symmetry constraints**.
3. Eliminating operations that would require **violating topology**.

This creates computation that is:

- parallel by construction,
 - safe by exclusion,
 - interpretable through geometry.
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6.3 Processor Model (SCFSG CPU)

The Symmetry-Constrained Finite-State Geometry (SCFSG) processor implements:

- spatial registers,

- adjacency-driven execution,
- constraint enforcement without global logic masking.

The instruction model derives from:

- allowable transitions,
- formation-stabilizing equilibria,
- topology-generated truth sets.

6.4 Educational and Physical Embodiments

Physical devices may include:

- interlocking polyhedral assemblies,
- tactile representation of spatial logic,
- hands-on demonstrations of **illegal state prevention**.

7. Advantages

Traditional Computation SCSC Advantages

Illegal but masked states Illegal states **cannot exist**

Symbol first	Structure first
High energy cost	Constraint enforced by geometry
Brittle AI abstractions	Categories emerge from invariants
Limited interpretability	Transparent spatial reasoning

8. Claims (Broad Foundational Claims)

Claim 1

A computation system in which digital state representations are encoded as **spatially constrained formations**, wherein allowable logical transitions occur exclusively along predefined **adjacency pathways** that preserve the symmetry of the state space.

Claim 2

A logic device comprising **volumetric logic points** defined by polyhedral structures that inherently prevent illegal states by **geometric exclusion**, the device configured to compute through structural transformation rather than symbolic processing.

Claim 3

A processor architecture (SCFSG) implementing **symmetry-constrained finite-state geometry**, wherein computation comprises simultaneous evaluation of all valid spatial transitions without any global instruction masking.